HIC - MID Term.

1. formative 1 2
2. formative 2 14
3. Quiz 1 HIC 33
4. Assignment 1 38
5. Concept of health Questions 45
6. meningits Questions 55
7. Leprosy Questions 59
8. Meningitis case GD 64

"اللهم صل وسلم و بارك على سيدنا محمد"

دعواتكم بظهر
الغيب في هذه
الأيام المفترجة

Formative 1 answered

1	**Which stage of disease progression involves the highest severity of signs and symptoms?**
A	Decline stage with reduced pathogen load
B	Incubation stage with silent pathogen growth
C	Convalescence stage with gradual recovery
D	Illness stage with pronounced clinical features

2	**A newborn receives maternal antibodies through the placenta, protecting him against infections during the first months of life. Which type of immunity is this?**		
A	Artificial immunity	B	Specific resistance
C	Passive immunity	D	Active immunity

3	**Which definition best distinguishes disease?**
A	A social inability to maintain daily roles
B	A mild subjective discomfort without signs
C	A physiological or psychological dysfunction
D	A temporary response to emotional stress

4	**In typhoid, the time during which the organism can be transmitted is known as what?**		
A	Period of communicability	B	Virulence
C	Pathogenicity	D	Secondary attack rate

5	**Which stage of natural disease history occurs before symptom onset but after pathological changes begin?**
A	Convalescence period after active illness
B	Decline period during pathogen elimination
C	Preclinical stage representing biological onset
D	Prodromal stage with nonspecific symptoms

6	**A man is bitten by a rabid dog and develops rabies; what direct transmission mode is involved?**		
A	Soil contact	B	Vertical transmission
C	Droplet infection	D	Contact with animals

7	**Which statement defines illness?**
A	A permanent decline from normal health
B	A subjective feeling of not being well
C	A socially assigned disability state
D	A measurable biological malfunction

8	**Case Fatality Ratio (CFR) is a good measure of?**						
A	Pathogenicity	B	Infectivity	C	Virulence	D	Infectiousness

9	**Which period is typically associated with maximum clinical severity?**		
A	Decline period	B	Incubation period
C	Convalescence period	D	Illness period

10	**A student with mumps is isolated; his apparently healthy brother is kept at home for two weeks. What is the brother's carrier state?**						
A	Temporary	B	Convalescent	C	Incubatory	D	Healthy

11	**Which statement best reflects the spectrum of health?**
A	Illness always indicates irreversible decline
B	Health ranges from complete wellness to death
C	Disease and health exist as fixed categories
D	Health begins only when symptoms appear

12	**Malaria parasite multiplies and develops inside a mosquito; which type of transmission is this?**		
A	Cyclodevelopmental	B	Propagative
C	Cyclopropagative	D	Mechanical

13	**A patient receives hepatitis B vaccine produced using recombinant DNA technology. Which type of vaccine is this?**		
A	Recombinant	B	Inactivated
C	Live attenuated	D	Subunit

14	**A patient inhales droplets 5–10 µm from a measles case; what type of transmission is this?**		
A	Vehicle	B	Droplet infection
C	Air borne	D	Vector

15	**A parasite develops but does not multiply inside a vector; what type of biological transmission is this?**		
A	Mechanical	B	Cyclodevelopmental
C	Propagative	D	Cyclopropagative

16	**The occurrence of health or disease depends on which interaction among?**
A	Agent, environment and host
B	Exposure, induction and disease
C	Time, place and person
D	Implementation, monitoring and evaluation

17	**A patient receives chloroquine to prevent malaria. Which type of prophylaxis is this?**		
A	Chemoprophylaxis	B	Seroprophylaxis
C	Active immunization	D	Passive immunization

18	**A community living near industrial pollution reports increased respiratory illnesses. Which determinant of health is most implicated?**
A	Cultural environment based on local traditions
B	Genetic predispositions with inherited abnormalities
C	Physical environment associated with air quality exposure
D	Social networks with variable interpersonal relations

19	**A child receives Salk polio vaccine prepared from killed organisms. Which type of vaccine is this?**		
A	Recombinant	B	Subunit
C	Live attenuated	D	Inactivated

20	**Which period of disease is characterized by pathogen multiplication with vague symptoms?**		
A	Decline stage	B	Illness stage
C	Prodromal stage	D	Convalescence stage

21	**A reconstituted measles vaccine is left unused for several hours. What happens?**		
A	Potency increased	B	Potency unaffected
C	Potency maintained	D	Potency lost

22	**A hospital sterilizes surgical instruments to prevent infections. Which preventive measure is this?**		
A	Secondary prevention	B	Rehabilitation
C	Specific protection	D	Health promotion

23	**Infected dust >10 µm commonly spreads which disease in hospitals?**						
A	Smallpox	B	Tuberculosis	C	Measles	D	Influenza

24	**Which dimension of health is most closely reflected by self-control and stable emotional responses?**
A	Vocational dimension emphasizing economic needs
B	Emotional dimension based on response regulation
C	Social dimension emphasizing group relationships
D	Mental dimension focusing on cognitive processes

25	**A housefly transfers typhoid organisms from feces to food via its legs; what type of vector is it?**		
A	Cyclopropagative	B	Biological
C	Propagative	D	Mechanical

26	**A cataract patient receives intraocular lens implantation to restore vision. Which aspect of prevention is this?**		
A	Secondary prevention	B	Primary prevention
C	Disability limitation	D	Rehabilitation

27	**Which determinant best explains why lower education levels correlate with poor health outcomes?**
A	Improved awareness of community disease patterns
B	Enhanced access to diagnostic and curative facilities
C	Guaranteed employment regardless of skill set
D	Increased stress and reduced confidence in health decisions

28	**Which factor belongs to the environmental component of the triad?**
A	Psychological resilience of individuals
B	Genetic predisposition to illness
C	Biological dose of the pathogen
D	Physical conditions where the host lives

29	**A patient with advanced cancer receives aggressive treatment to prevent further disability. Which level of prevention is this?**		
A	Secondary prevention	B	Tertiary prevention
C	Rehabilitation	D	Primary prevention

30	**A person develops cutaneous anthrax after handling contaminated wool; what is the portal of entry?**						
A	Skin	B	Conjunctival	C	GI tract	D	Respiratory

31	**Which determinant explains population health patterns beyond individual control?**
A	Personal preference for certain food types
B	Individual desire for physical activity
C	Contextual social and economic environment
D	Daily choice of recreational hobbies

32	**Which statement reflects the concept of well-being?**
A	A continuous state of economic success
B	A lack of biological abnormalities
C	A positive judgment of life and feeling good
D	A clinical absence of disease

33	**Which element is an example of a social determinant of health?**		
A	Number of daily exercise sessions	B	Personal taste preferences
C	Red blood cell count levels	D	Income and social status

34	**Which component of the epidemiological triad refers to host-related characteristics?**		
A	Environmental factors	B	Agent factors
C	Host factors	D	Social behavior factors

35	**Infectious organisms present on utensils represent what concept?**						
A	Infection	B	Colonization	C	Infestation	D	Contamination

36	**A patient receives human rabies immunoglobulin. Which type of prophylaxis is this?**		
A	Chemoprophylaxis	B	Active immunization
C	Passive natural immunity	D	Seroprophylaxis

37	**Which health dimension includes the ability to maintain relationships and integrate with society?**
A	Vocational dimension describing work capability
B	Spiritual dimension describing philosophical beliefs
C	Physical dimension describing organ performance
D	Social dimension describing interpersonal functioning

38	**An infected person who shows no symptoms but remains infectious for life is what type of carrier?**						
A	Incubatory	B	Chronic	C	Healthy	D	Convalescent

39	**Cholera transmitted through contaminated water represents which route?**						
A	Vehicle	B	Vector	C	Air borne	D	Fomite

40	**A young couple seeks premarital counseling to avoid genetic disorders in their offspring. Which preventive measure is being applied?**		
A	Health education	B	Disability limitation
C	Genetic counseling	D	Rehabilitation

41	**Which determinant explains why employed people often have better health outcomes?**
A	Access to entertainment facilities
B	Having predictable daily routines
C	Working longer hours without breaks
D	Better working conditions and job control

42	**A person exposed to influenza but protected due to existing antibodies is showing what factor?**		
A	Communicability	B	Virulence
C	Pathogenicity	D	Host immunity

43	**A patient inhales droplet nuclei <5 µm from a TB case; what mode of transmission is this?**		
A	Vehicle transmission	B	Fomite transmission
C	Air borne transmission	D	Droplet infection

44	**A worker recovering from a back injury is given modified work conditions to safely return to employment. Which preventive measure is this?**		
A	Rehabilitation	B	Tertiary prevention
C	Secondary prevention	D	Primary prevention

45	**A child with mild respiratory infection is scheduled for vaccination. Should vaccination be postponed?**		
A	Yes, only for OPV	B	No, not contraindicated
C	Yes, only for BCG	D	Yes, contraindicated

46	**A food handler remains infectious with typhoid for years without symptoms; what type of carrier is he?**						
A	Incubatory	B	Convalescent	C	Chronic	D	Temporary

47	**A new influenza strain spreads across continents; what level of occurrence is this?**						
A	Endemic	B	Pandemic	C	Epidemic	D	Outbreak

48	**A newborn acquires ophthalmia neonatorum while passing through an infected birth canal; what is the portal of entry?**		
A	Skin	B	Conjunctiva
C	GI tract	D	Respiratory tract

49	**Dengue virus multiplies inside Aedes mosquitoes; what type of transmission is this?**		
A	Propagative	B	Cyclodevelopmental
C	Cyclopropagative	D	Mechanical

50	**A patient with hearing loss is provided with a hearing aid to restore communication ability. This is an example of:**		
A	Specific protection	B	Disability limitation
C	Health promotion	D	Rehabilitation

51	**Which description characterizes the agent in the epidemiological triad?**
A	The cultural behaviors of communities
B	The surroundings affecting disease spread
C	The factor responsible for causing disease
D	The internal immunity of individuals

52	**How easily disease spreads among susceptible contacts is measured by which indicator?**		
A	Pathogenicity	B	Case fatality rate
C	Period of communicability	D	Secondary attack rate

53	**Soil is a reservoir for which disease?**						
A	Histoplasmosis	B	Measles	C	Rabies	D	Dengue

54	**A patient with impairment due to spinal injury is unable to perform routine activities. Which stage of disease progression is this?**						
A	Impairment	B	Handicap	C	Disability	D	Disease

55	**A patient recovering from stroke is enrolled in occupational therapy to regain independence. Which level of prevention is this?**		
A	Primary prevention	B	Tertiary prevention
C	Secondary prevention	D	Rehabilitation

56	**A patient recovering from influenza develops immunity that lasts only a short period. Which type of immunity is this?**						
A	Absolute	B	Moderate	C	Solid	D	Transient

57	**Plague bacilli multiply inside a rat flea; what is this?**		
A	Propagative transmission	B	Mechanical transmission
C	Cyclodevelopmental transmission	D	Cyclopropagative transmission

58	**A disease that produces lifelong immunity likely has which characteristic?**		
A	High virulence	B	Strong antigenic power
C	Weak antigenic power	D	Low pathogenicity

59	**Which factor most influences the length of an incubation period?**
A	Dose, site of infection, and host immunity level
B	Personal preferences, habits, and attitudes
C	Age, ethnicity, and family health history
D	Social networks, occupation, and culture

60	**A patient feels unwell, yet clinical exams show no abnormality. This state is best classified as:**
A	Sickness as social role impairment in society
B	Pathology as an advanced stage of organ damage
C	Disease as a measurable physiological dysfunction
D	Illness as a subjective experience of discomfort

61	**A child with neurological disorder should not receive which vaccine component?**						
A	Tetanus	B	Pertussis	C	Hepatitis B	D	Measles

62	**A child receives the measles vaccine and develops immunity without contracting the disease. Which type of immunity is this?**		
A	Passive immunity	B	Natural active immunity
C	Non-specific resistance	D	Artificial active immunity

63	**Which physical environment factor contributes positively to health?**
A	Access to safe water and clean air
B	Residence in overcrowded housing
C	Work in hazardous industrial sites
D	Constant exposure to excessive noise

64	**A child receives vitamin A supplementation at 18 months. Which dose is given?**						
A	200,000 IU	B	100,000 IU	C	50,000 IU	D	300,000 IU

65	**A patient with tuberculosis is treated promptly to reduce illness duration and prevent spread in the community. Which principle of prevention is applied?**
A	Health promotion
B	Early diagnosis and prompt treatment
C	Rehabilitation
D	Disability limitation

66	**Which component of the epidemiological triad involves characteristics of the disease-causing entity?**
A	Environmental factors describing physical conditions
B	Behavioral factors describing coping strategies
C	Host factors describing personal susceptibility
D	Agent factors describing pathogen characteristics

67	**Which phase of natural disease history involves biological changes without symptoms?**		
A	Outcome phase	B	Recurrence phase
C	Clinical phase	D	Preclinical phase

68	**A patient with impairment in leg function is unable to walk, resulting in disability. Which stage of disease progression does this represent?**						
A	Impairment	B	Disease	C	Handicap	D	Disability

69	**A patient receives influenza vaccine prepared from a viral component. Which type of vaccine is this?**						
A	Subunit	B	Recombinant	C	Toxoid	D	Inactivated

70	**A 45-year-old woman undergoes a mammogram to detect breast cancer before symptoms appear. This is an example of:**		
A	Secondary prevention	B	Tertiary prevention
C	Rehabilitation	D	Primary prevention

71	**Which definition best reflects the WHO concept of health?**
A	A personal perception of happiness and satisfaction
B	A simple absence of disease and clinical infirmity
C	A complete state of physical, mental, and social well-being
D	A stable physiological state with minimal pathology

72	**A newborn receives BCG vaccine at birth. Which route is used?**						
A	Oral	B	SC	C	IM	D	ID

73	**A patient receives rabies vaccine prepared in human diploid cells. Which type of vaccine is this?**		
A	Recombinant vaccine	B	Tissue culture vaccine
C	Subunit vaccine	D	Inactivated vaccine

74	**A community health program emphasizes nutrition education, personal hygiene, and lifestyle modification to strengthen individuals against disease. Which type of intervention does this represent?**		
A	Specific protection	B	Disability limitation
C	Health promotion	D	Rehabilitation

75	**A patient recovering from amoebiasis continues to shed cysts; what type of carrier is this?**						
A	Convalescent	B	Chronic	C	Temporary	D	Incubatory

76	**A patient with advanced disability is given vocational training to restore earning capacity. Which aspect of rehabilitation is this?**		
A	Psychological rehabilitation	B	Vocational rehabilitation
C	Social rehabilitation	D	Physical rehabilitation

77	**Leptospirosis consistently maintained by rodents in a region represents what?**						
A	Sporadic	B	Epidemic	C	Pandemic	D	Endemic

78	**The ability of a microorganism to cause disease is called what?**		
A	Virulence	B	Communicability
C	Pathogenicity	D	Antigenic power

79	**A country with constant low-level malaria transmission shows what pattern?**						
A	Sporadic	B	Epidemic	C	Endemic	D	Pandemic

80	**A school introduces daily physical activity and healthy eating habits for all students, regardless of their current health status. Which strategy of primordial prevention is being applied?**		
A	High-risk strategy	B	Disability limitation
C	Population strategy	D	Rehabilitation strategy

Formative 2

1	**Which of the following is the best method for disposal of hospital refuse such as syringes and needles?**		
A	Incineration	B	Burial in soil
C	Washing with water	D	Storing in sealed bags

2	**In a densely populated urban district, tuberculosis incidence has remained stable for 5 years despite good treatment coverage. Which factor most likely explains the continued transmission?**
A	Low case fatality rate
B	High proportion of extrapulmonary TB cases
C	Delayed diagnosis of infectious pulmonary cases
D	High BCG vaccination coverage

3	**Which clinical symptom combination most strongly suggests pulmonary TB?**
A	Chronic cough above 3 weeks with fever and weight loss
B	Persistent headache with photophobia and vomiting
C	Acute diarrhea with rapid dehydration and cramps
D	Sudden joint pain with localized swelling and redness

4	**Which preventive measure offers partial protection against leprosy?**		
A	MMR vaccine	B	BCG vaccine
C	Polio vaccine	D	Hepatitis B vaccine

5	**A public health survey finds that tuberculosis prevalence is highest in low-income urban settlements. Which determinant most strongly explains this distribution?**		
A	Seasonal climate changes	B	Genetic susceptibility
C	High vaccination coverage	D	Overcrowding and poor ventilation

6	**Which vaccine type offers improved immunogenicity and herd protection against meningococcal disease?**
A	Conjugate vaccines targeting A, C, W-135, Y
B	DNA-based vaccines targeting multiple serotypes
C	Live attenuated vaccines targeting serogroup B
D	Polysaccharide vaccines with lifelong protection

7	**In developing countries, the prevalence of HAIs is significantly higher than in developed countries. What is the approximate prevalence range in developing countries?**						
A	50–60%	B	15–20%	C	25–40%	D	7–12%

8	**Which environmental control is most effective during meningitis epidemics?**
A	Encouraging outdoor exposure during peak hours
B	Increasing humidity in enclosed environments
C	Reducing access to shared recreational facilities
D	Improving ventilation in crowded institutions

9	**Which statement describes the role of isolation during meningococcal epidemics?**
A	It has limited effect because carriers are more significant sources
B	It replaces the requirement for vaccination campaigns
C	It is essential because clinical cases transmit the majority
D	It eliminates the need for chemoprophylaxis programs

10	**Which regimen is recommended for preventive therapy in TB contacts?**
A	Monthly streptomycin injections
B	Daily isoniazid at adult dose of 300 mg
C	Weekly rifampicin at minimal dosing
D	Quarterly ethambutol dosing

11	**Which leprosy reaction is commonly seen in lepromatous leprosy?**		
A	Reversal reaction	B	Erythema nodosum leprosum (ENL)
C	Type IV hypersensitivity	D	Anaphylactic reaction

12	**Which population serves as the main reservoir for M. leprae?**						
A	Rodents	B	Humans	C	Soil bacteria	D	Birds

13	**A close contact of an infectious TB case tests negative on PPD during exposure and returns after 10 weeks for repeat testing. Why is this necessary?**
A	To verify improvement of physical symptoms post-exposure
B	To assess early lung damage following acute exposure
C	To detect delayed hypersensitivity conversion after exposure
D	To confirm eradication of infection by immune system

14	**Which factor most significantly affects TB transmission probability?**
A	Type of food and nutritional habits of contacts
B	Genetic similarities between index and contacts
C	Age, gender, and education of exposed persons
D	Infectiousness, environment, duration of exposure

15	**In a rural community, the prevalence of tuberculosis is decreasing but the incidence remains unchanged. Which interpretation is most appropriate?**
A	Population immunity is declining
B	Duration of disease is shortening due to improved treatment
C	Case detection is decreasing
D	Transmission is increasing

16	**How soon after beginning effective antibiotic therapy does a typical meningococcal case stop being infectious?**
A	After the completion of the entire drug regimen
B	Only after laboratory confirmation of sterile CSF
C	Within the first 24 hours of treatment initiation
D	Within the first 5 days after clinical improvement

17	**A patient develops infection after lumbar puncture due to poor aseptic technique. Which cause of HAI is most relevant?**
A	Drug resistance
B	Contaminated food
C	Overcrowding
D	Failure to observe aseptic precautions

18	**What is the primary mode of transmission of leprosy?**
A	Direct skin contact
B	Respiratory droplets during prolonged close exposure
C	Insect bites
D	Contaminated water

19	**A patient develops infection after being treated with contaminated wound dressings. Which transmission mechanism is most likely?**		
A	Droplet infection	B	Fomite transmission
C	Contact transmission	D	Dust infection

20	**Universal blood and body fluid precautions are primarily aimed at preventing transmission of which infections?**		
A	Escherichia coli and Klebsiella	B	Tuberculosis and malaria
C	HBV and HIV	D	Influenza and measles

21	**A hospital maintains 12 ft spacing between beds and ensures cross ventilation. Which transmission route is being controlled?**		
A	Contact transmission	B	Droplet transmission
C	Dust transmission	D	Fomite transmission

22	**Which finding best differentiates lepromatous from tuberculoid leprosy in a field epidemiology survey?**
A	Strong lepromin positivity
B	Multiple symmetrical lesions with nasal mucosa involvement
C	Localized lesions with sharp margins
D	Presence of fever

23	**Which mass chemoprophylaxis policy applies in meningitis epidemics?**
A	Recommended for all individuals in open community settings
B	Recommended solely for symptomatic clinical cases
C	Recommended only for closed, medically supervised communities
D	Recommended only after vaccination has been completed

24	**Over 60% of global deaths are attributed to NCDs. Which disease group contributes the highest mortality?**		
A	Digestive diseases	B	Musculoskeletal disorders
C	Cardiovascular diseases	D	Chronic respiratory diseases

25	**Which group is MOST vulnerable to HAIs?**		
A	Vaccinated individuals	B	Post-surgery patients
C	Hospital staff	D	Healthy adults

26	**Which modifiable risk factor contributes to approximately 3.3 million deaths annually worldwide?**		
A	Harmful use of alcohol	B	Insufficient physical activity
C	Tobacco use	D	High salt intake

27	**Which measure best reflects the burden of chronic diseases beyond mortality figures?**
A	Years of Potential Life Lost (YPLL)
B	Case fatality rate
C	Attack rate
D	Incidence rate

28	**A meningitis epidemic is declared in a refugee camp. Which public health measure is most appropriate?**
A	Replacing all camp healthcare staff immediately
B	Blocking all transportation routes to nearby towns
C	Enhancing ventilation and reducing crowding in shelters
D	Increasing surface disinfection of shared objects

29	**Which factor most contributes to rapid spread during meningitis epidemics?**
A	High humidity and frequent exposure to rainfall
B	High survival of the organism outside the human body
C	Minimal contact between individuals in communities
D	Overcrowded living environments with poor ventilation

30	**Which major component makes DOTS an effective TB control strategy?**
A	Sole reliance on radiographic screening programs
B	Exclusive treatment of smear-negative patients
C	Direct observation of therapy during initial months
D	Mandatory hospitalization of all infectious cases

31	**A region experiences a sudden increase in tuberculosis incidence after several years of decline. Which situation most likely contributed to this change?**
A	Decreased population density
B	Increase in immunocompromising conditions such as HIV/AIDS
C	Expansion of vaccination programs
D	Improved socioeconomic conditions

32	**Why is rifampicin considered the most crucial drug in multidrug therapy (MDT) for leprosy control?**
A	It rapidly kills M. leprae within days, reducing infectiousness
B	It prevents nerve thickening
C	It improves skin pigmentation
D	It is the cheapest drug

33	**What is the average incubation period for leprosy?**						
A	1 year	B	5 years	C	20 years	D	10 years

34	**A country reports that most new tuberculosis cases occur among young adults aged 20–40 years. Epidemiologically, this pattern most likely indicates:**
A	Reactivation of old infections in the elderly
B	Active and ongoing transmission in the community
C	Effective childhood vaccination programs
D	Declining TB transmission

35	**Which major risk period follows TB infection and carries the highest likelihood for clinical disease development?**
A	First 6–12 months following initial infection
B	Second decade of life for all age groups
C	Final months of convalescence after therapy
D	Fifth to eighth year after initial exposure

36	**Which clinical feature is most strongly associated with early invasive meningococcal disease progression?**
A	Chronic fatigue with reduced appetite
B	Gradual muscle weakness over several weeks
C	Persistent dry cough with low-grade fever
D	Severe headache with photophobia and vomiting

37	**During contact investigation of a smear-positive tuberculosis patient, several household contacts are found to have latent infection but no active disease. Which control measure is most appropriate to prevent future cases?**		
A	Isolation for 2 weeks	B	Preventive chemotherapy
C	Repeat sputum microscopy monthly	D	Immediate hospitalization

38	**A university dormitory reports several students with nasopharyngitis followed by one suspected meningitis case. What is the most important mode of transmission to consider?**
A	Fomite contamination through shared equipment
B	Vector-borne spread from local insect populations
C	Waterborne exposure in shared sanitation areas
D	Airborne spread through prolonged close contact

39	**Which action is most vital during meningitis epidemic control efforts?**
A	Delaying treatment until laboratory confirmation
B	Minimizing surveillance to reduce public concern
C	Replacing antibiotics with environmental modification
D	Early case detection and prompt initiation of treatment

40	**Which serogroups of Neisseria meningitidis are most commonly associated with invasive disease worldwide?**						
A	B, C, X, Y	B	A, E29, X, Y	C	A, B, C, W-135	D	W-135, X, Y, Z

41	**A hospital outbreak of Klebsiella infections is traced back to contaminated IV fluids. Which transmission mechanism is most likely?**		
A	Contact transmission	B	Dust infection
C	Iatrogenic transmission	D	Droplet infection

42	**A patient with untreated multibacillary leprosy lives with several family members. Which group is most at risk of developing infection?**
A	People living in neighboring villages
B	Household contacts with prolonged exposure
C	Health workers using protective measures
D	Casual visitors

43	**Which administrative measure is essential for outbreak detection and control of HAIs?**
A	Hospital Infection Control Committee
B	Concurrent disinfection
C	Barrier nursing
D	Vaccination of patients

44	**A hospital staff member with influenza continues working and infects multiple patients. Which preventive measure was neglected?**		
A	Barrier nursing	B	Concurrent disinfection
C	Staff with infections off duty	D	Task nursing

45	**Raised cholesterol is most strongly associated with which outcome?**		
A	Chronic respiratory disease	B	Stroke and heart disease
C	Musculoskeletal disorders	D	Diabetes

46	**A region reports low prevalence but continues active surveillance and early treatment programs. What is the most important reason for maintaining these measures?**
A	To shorten treatment duration
B	To increase hospital admissions
C	To eliminate all bacterial diseases
D	To prevent reintroduction and continued transmission

47	**In a hospital where overcrowding is common and aseptic precautions are poorly followed, which factor most strongly predisposes to HAIs?**		
A	Drug resistance in organisms	B	Adequate ventilation
C	Vaccination of staff	D	Proper sterilization of equipment

48	**During disaster or war situations, interruption of leprosy treatment programs may lead to which major public health consequence?**
A	Immediate eradication of the disease
B	Shorter incubation period
C	Reduction in new infections
D	Increased transmission and complications

49	**Which screening measure is routinely recommended for all close TB contacts?**
A	PPD tuberculin skin testing for initial evaluation
B	Routine blood culture to detect bacteremia
C	Mandatory CT scan for early lung imaging
D	Immediate sputum culture for every exposed person

50	**A patient admitted for pneumonia develops urinary tract infection after catheterization. Which source of infection is most likely?**		
A	Contaminated food	B	Environmental reservoir
C	Cross-infection from staff	D	Patient's own flora

51	**Which indirect cost is commonly associated with chronic diseases?**		
A	Productivity loss	B	Medication expenses
C	Hospital bills	D	Surgery costs

52	**Why might borderline leprosy shift toward the lepromatous end of the spectrum if untreated?**
A	Due to reinfection with a different strain
B	Because it is immunologically unstable with declining cell-mediated immunity
C	Due to improved immunity
D	Due to failure of B-cell activation

53	**A community health worker identifies a patient with hypopigmented skin lesions and loss of sensation. Early treatment is started immediately. What is the most important public health benefit of this action?**
A	Prevention of nerve damage and disability
B	Shortening of the incubation period
C	Elimination of environmental sources of infection
D	Reduction of stigma in the community

54	**A tuberculosis control program identifies that most transmission occurs in overcrowded households. Which primary prevention strategy would most effectively reduce this risk?**
A	Expansion of BCG vaccination programs
B	Increasing antibiotic prophylaxis for the population
C	Routine mass chest X-ray screening
D	Early detection and treatment of infectious cases

55	**During a meningitis epidemic, which indicator most reliably signals that the outbreak is intensifying?**
A	A consistent reduction in severe clinical cases
B	A stable carrier level across multiple months
C	A gradual decline in reported nasopharyngeal cases
D	A progressive rise in asymptomatic carrier rates

56	**A contact has a positive PPD but normal chest X-ray. What is the appropriate next step?**
A	Request repeated sputum smear daily
B	Begin isoniazid preventive treatment
C	Delay therapy until symptoms appear
D	Start full four-drug TB treatment

57	**Which environmental condition most strongly contributes to meningococcal meningitis outbreaks in the African meningitis belt?**
A	Cold and dry seasonal winds with dust exposure
B	Hot and humid equatorial rainfall patterns
C	Stable year-round temperatures with high moisture
D	Irregular monsoon cycles with mild variations

58	**What is the first-line drug for meningococcal meningitis treatment?**
A	Rifampicin used for both cases and contacts
B	Ceftriaxone reserved exclusively for prophylaxis
C	Ciprofloxacin given as a short-course regimen
D	Penicillin administered promptly on diagnosis

59	**Which group is most at risk among contacts of infectious TB cases?**
A	Occasional visitors with minimal direct interactions
B	Casual community members without close proximity
C	Close household contacts with prolonged daily exposure
D	Distant neighbors with outdoor-only interactions

60	**During a screening campaign, a patient presents with a single well-defined hypopigmented lesion with marked sensory loss and strong cell-mediated immunity. Which type of leprosy is most likely?**		
A	Lepromatous leprosy	B	Tuberculoid leprosy
C	Multibacillary leprosy	D	Borderline leprosy

61	**A nurse forgets to disinfect her hands after handling a patient's sputum and then attends another patient. Which transmission route is most likely?**		
A	Fomite transmission	B	Dust infection
C	Contact transmission	D	Droplet infection

62	**In a village with several untreated multibacillary leprosy cases, new infections continue to appear among close household contacts. Which factor most strongly explains this pattern?**
A	Seasonal climate changes
B	Contaminated water supply
C	Prolonged close exposure to infectious untreated cases
D	Genetic susceptibility of villagers

63	**Which contacts should receive preventive therapy even if initially PPD-negative?**
A	Elderly individuals without recent illness history
B	Individuals with occasional short-term exposures
C	Children under five years or HIV-infected individuals
D	Adults with stable immune systems and no symptoms

64	**A household contact tests negative on a PPD immediately after exposure. What is the next correct step?**
A	Repeat the PPD after 2–3 months for possible conversion
B	Begin full TB treatment without further evaluation
C	Discontinue follow-up due to low transmission risk
D	Initiate immediate chest radiography every week

65	**A national tuberculosis control program prioritizes early detection of infectious cases. Which group should be targeted first for screening to most effectively interrupt transmission?**
A	Patients with extrapulmonary tuberculosis
B	Patients who completed treatment years ago
C	Individuals previously vaccinated with BCG vaccine
D	Household contacts of smear-positive tuberculosis cases

66	**During an inter-epidemic period in a Sub-Saharan African region, surveillance shows a carrier rate of 30%. What does this indicate epidemiologically?**
A	A moderate baseline that may rise before potential epidemics
B	A stable prevalence with low likelihood of immediate outbreaks
C	A gradual decline toward post-epidemic recovery phase
D	A negligible level insufficient for transmission dynamics

67	**A meningitis outbreak occurs in a military training base. What is the most appropriate measure?**
A	Replace routine treatment with serologic screening only
B	Restrict antibiotic use only to confirmed clinical cases
C	Postpone all surveillance activities during the outbreak
D	Implement mass chemoprophylaxis within the base

68	**What cardinal sign confirms the diagnosis of leprosy in endemic areas?**
A	Skin lesion with definite sensory loss
B	Positive lepromin test
C	Leukocytosis
D	Elevated ESR

69	**Which statement about prevention of chronic diseases is correct?**
A	Genetic risks cannot be modified
B	Only 20% are preventable
C	Prevention is less effective than treatment
D	80% are preventable with existing measures

70	**Which diagnostic test can confirm TB before it becomes symptomatic?**
A	Tuberculin skin test indicating latent infection
B	Blood culture indicating systemic dissemination
C	Chest radiograph indicating early cavitation only
D	Sputum smear indicating only advanced disease

71	**Which preventive measure should be strengthened during early meningococcal epidemic detection?**
A	Immediate nationwide vaccination without targeting
B	Daily testing of schoolchildren for meningococcal bacteria
C	Surveillance to identify rises in nasopharyngeal carriage
D	Mass treatment of all reported upper respiratory infections

72	**A patient develops permanent disability after a spinal injury. Which characteristic of chronic disease does this illustrate?**		
A	Residual disability	B	Temporary impairment
C	Short-term supervision	D	Reversible pathology

73	**In a control program, emphasis is placed on multidrug therapy rather than single-drug treatment. What is the main reason for using combination therapy?**
A	To shorten incubation period
B	To reduce treatment cost
C	To prevent development of drug resistance
D	To improve vaccine effectiveness

74	**Which statement about vaccination during meningitis epidemics is correct?**
A	Vaccination is contraindicated in all adolescent groups
B	Conjugate vaccines covering A, C, W-135, Y are protective
C	Polysaccharide vaccines offer lifelong herd immunity
D	Vaccination replaces the need for chemoprophylaxis

75	**What is the main purpose of chest radiography among PPD-positive contacts?**
A	To detect possible evolution into active TB disease
B	To confirm reinfection from unrelated sources
C	To evaluate environmental causes of dry cough
D	To determine nutritional status and body mass

76	**What public health consequence may occur if MDT supply is interrupted during disasters or conflict?**
A	Increased transmission, complications, and drug resistance
B	Rise in asymptomatic infections only
C	Reduced case detection
D	Spontaneous cure of cases

77	**Which factor determines the need for extended follow-up among contacts?**
A	Financial status, occupation, and education level
B	Cultural background, ethnicity, and travel history
C	Duration of exposure, infectiousness, and immune status
D	Residence size, climate, and dietary preferences

78	**Which programmatic measure supports compliance among high-risk TB contacts?**
A	Exclusive reliance on radiographic monitoring
B	Occasional follow-up only after symptoms appear
C	Directly observed preventive therapy when adherence is uncertain
D	Self-reported adherence without clinical supervision

79	**Which clinical form of leprosy shows strong cell-mediated immunity?**						
A	Indeterminate	B	Borderline	C	Lepromatous	D	Tuberculoid

80	**In a community where tuberculosis incidence is declining, which epidemiological indicator would best demonstrate successful tuberculosis control?**
A	Sustained reduction in incidence of smear-positive tuberculosis
B	Increased number of TB suspects examined
C	Increased BCG vaccination coverage
D	Decreased case fatality rate among treated patients

81	**Which public health strategy is most effective in reducing community transmission of leprosy?**
A	Mass prophylactic antibiotics
B	Routine lepromin testing
C	Early detection and prompt initiation of MDT
D	Isolation of all patients

82	**A public health officer emphasizes health education about the curability of leprosy and availability of free treatment. What major public health problem is this strategy mainly addressing?**
A	Drug toxicity
B	Genetic susceptibility
C	Social stigma and delayed case detection
D	Environmental contamination

83	**Which type of TB is most commonly encountered among reported cases?**
A	Congenital TB transmitted during perinatal period
B	Avian-type TB transmitted from poultry exposure
C	Extra-pulmonary TB including skeletal involvement only
D	Pulmonary TB including smear-positive and smear-negative

84	**What is the recommended duration for standard preventive therapy using isoniazid?**
A	Weekly dosing for immune stimulation only
B	Daily dosing for sufficient long-term protection
C	Quarterly dosing for minimal medication exposure
D	Monthly dosing for stable long-term immunity

85	**Which virus is associated with liver cancer and can be prevented by vaccination?**						
A	HBV	B	HCV	C	HIV	D	HPV

86	**An outbreak investigation identifies a meningococcal carrier rate of 75% among a school population. Which public health action is appropriate?**
A	Avoid prophylaxis and rely on passive environmental surveillance
B	Recommend mass chemoprophylaxis for entire open communities
C	Implement mass chemoprophylaxis only in closed supervised groups
D	Initiate targeted chemoprophylaxis for close contacts only

87	**A 60-year-old man with diabetes and hypertension is advised to reduce salt intake. What is the recommended daily limit?**						
A	1 teaspoon	B	3 teaspoons	C	2 teaspoons	D	No restriction

88	**Which group has the highest risk of rapid progression from TB infection to TB disease?**
A	Travelers without history of previous infections
B	Children older than 12 years with mild symptoms
C	Adults with stable immune systems and low exposure
D	Patients living with HIV due to weakened immunity

89	**Which statement best reflects the global epidemiology of tuberculosis?**
A	One-third of the world population is infected with TB
B	Less than 1% of global TB patients develop complications
C	MDR-TB is not considered a public health emergency
D	TB causes fewer deaths than most acute viral infections

90	**Which organism causes leprosy?**		
A	Mycobacterium leprae	B	Mycobacterium tuberculosis
C	Mycobacterium kansasii	D	Mycobacterium bovis

91	**A contact of a leprosy patient is examined annually for several years even though no symptoms appear. What is the main reason for prolonged follow-up?**
A	Patients develop lifelong immunity immediately
B	The disease has a long and variable incubation period
C	The vaccine requires repeated monitoring
D	The infection spreads through vectors

92	**Which environmental risk factor is LEAST likely to contribute to NCDs?**		
A	Occupational hazards	B	Seasonal influenza
C	Water pollution	D	Air pollution

93	**A national tuberculosis program implements the Directly Observed Treatment Short-course (DOTS) strategy. Which epidemiological outcome is most directly expected?**
A	Increase in latent TB infection
B	Reduction in BCG vaccination coverage
C	Increase in TB transmission
D	Reduction in treatment default and drug resistance

94	**A 30-year-old sedentary office worker develops obesity and hypertension. Which risk factor is most relevant?**		
A	Ethnicity	B	Insufficient physical activity
C	Genetic predisposition	D	Age

95	**Which criterion must be met before beginning preventive therapy for TB contacts?**
A	Confirm latent infection by sputum culture
B	Require hospitalization for baseline assessment
C	Rule out active TB through history and chest X-ray
D	Perform mandatory CT scan for lung mapping

96	**Which WHO treatment regimen is recommended for multibacillary leprosy?**
A	Dapsone alone for 1 year
B	Rifampicin monthly only
C	Rifampicin + Dapsone + Clofazimine for 24 months
D	Rifampicin + Dapsone for 6 months

97	**Which financial burden statement is correct regarding chronic diseases?**
A	Death costs more than living with disease
B	Living with chronic disease is more expensive
C	Medicare spending is minimal on chronic diseases
D	Direct costs always exceed indirect costs

98	**For how long should household contacts of leprosy patients be followed up annually?**						
A	3 years	B	10 years	C	1 year	D	5 years

99	**A patient with lepromatous leprosy develops painful red nodules and systemic symptoms during treatment. Which complication is most likely responsible?**		
A	Reversal reaction	B	Septicemia
C	Acute allergic dermatitis	D	Erythema nodosum leprosum

Quiz 1 HIC الكل اختيار شاهي Answer

1	**According to WHO, health is defined as:**
A	Absence of disease only
B	Ability to work productively
C	Complete physical, mental, and social well-being
D	Normal laboratory findings

2	**Which is not a dimension of health in the WHO definition?**						
A	Emotional	B	Social	C	Mental	D	Physical

3	**A person who manages temper and controls tension is demonstrating:**		
A	Spiritual health	B	Social health
C	Emotional health	D	Vocational health

4	**Social health refers to:**		
A	Weight and height balance	B	Clear thinking
C	Moral values	D	Ability to form relationships

5	**Vocational health means:**		
A	High spirituality	B	Absence of illness
C	Good emotions	D	Ability to earn sufficiently

6	**The lowest point on the health-disease spectrum is:**		
A	Positive health	B	Death
C	Illness	D	Disability

7	**Well-being includes:**
A	Only emotional satisfaction
B	High intelligence
C	Absence of disease
D	Positive emotions and life satisfaction

8	**A determinant strongly linked with better health is:**		
A	Gender	B	Low education
C	Higher income	D	Lack of social support

9	**Disease is defined as:**
A	Physiological or psychological dysfunction
B	A feeling of being unwell
C	A productive lifestyle
D	Social malfunction

10	**Illness refers to:**
A	Social inability to work
B	Agent-host interaction
C	Objective dysfunction
D	Subjective feeling of not being well

11	**The epidemiologic triad includes all except:**						
A	Vector	B	Agent	C	Environment	D	Host

12	**The incubation period is the interval between:**		
A	Treatment and cure	B	Exposure and first symptom
C	Symptoms and diagnosis	D	Recovery and relapse

13	**The prodromal stage is characterized by:**		
A	Nonspecific early symptoms	B	Pathogen elimination
C	No symptoms	D	Maximum severity of symptoms

14	**The period of illness is when:**		
A	Recovery begins	B	Signs and symptoms are most severe
C	Symptoms are mild	D	Host-agent balance is stable

15	**During the decline stage, patients are at risk of:**		
A	Preclinical transmission	B	Secondary infections
C	Genetic mutation	D	High infectivity

16	**The convalescence period ends when:**
A	Pathogen load increases
B	The patient returns to normal function
C	Symptoms peak
D	No damage remains

17	**Which determinant is most influenced by government policy?**		
A	Gender	B	Genetics
C	Physical environment	D	Age

18	**A person unable to perform a social role is considered:**						
A	Sick	B	Diseased	C	Ill	D	Disabled

19	**Health and disease are considered:**		
A	A dynamic process	B	Determined only by biology
C	Binary states	D	Completely unrelated

20	**Psychological well-being is closest to:**		
A	Spiritual health	B	Social health
C	Emotional health	D	Mental health

21	**Which is most accurate regarding the spectrum of health?**
A	Illness is a distinct boundary separate from wellness
B	Movement along the spectrum is gradual and continuous
C	Individuals stay in one fixed state
D	Death is not part of the spectrum

22	**Which determinant combination best explains health inequity?**		
A	Income and education	B	Water quality and age
C	Gender and genetics	D	Behavior and sex

23	**A pathogen with a long incubation period challenges public health because:**
A	It cannot cause outbreaks
B	It spreads silently before symptoms appear
C	It causes mild disease
D	It is easily detected early

24	**Which component of the epidemiologic triad can be modified fastest in an outbreak?**		
A	Host genetics	B	Host age distribution
C	Agent virulence	D	Environment

25	**In the natural history of disease, early detection programs target which stage?**						
A	Preclinical	B	Prodromal	C	Clinical	D	Incubation

26	**Which period is most appropriate for isolation to prevent peak transmission?**						
A	Decline	B	Illness period	C	Convalescence	D	Recovery

27	**Disability as an outcome of disease indicates:**		
A	Preclinical infection	B	Full return to well-being
C	Pathogen eradication	D	Remaining pathology after recovery

28	**Spiritual dimension of health helps individuals by:**
A	Preventing all diseases
B	Improving body measurements
C	Replacing medical treatment
D	Providing meaning and high-level thinking

29	**During the prodromal stage, public health focus is:**
A	Identifying symptoms early to prevent spread
B	Isolation only
C	Administering surgery
D	Declaring epidemic status

30	**Which factor best differentiates illness from disease?**		
A	Subjective experience	B	Laboratory confirmation
C	Agent virulence	D	Social role performance

Answer the following question

1. Define:

- Epidemiology

The study of distribution and determinants of health-related states or events in specified populations and application of this study to the control of health problems.

- Endemic

Constant presence or usual prevalence of a disease in a population/geagraphic area.

- Epidemic

Sudden increase in cases above what is normally expected in a population/area.

- Infectious Disease.

Disease directly resulting from infection.

- Contagious Disease

Spread person-to-person via close physical contact (e.g. scabies, STDs).

- Communicable Disease

Illness due to infectious agent/toxin, transmitted directly or indirectly from reservoir to host.

- Zoonotic Disease (Zoonosis)

Disease naturally transmitted between vertebrate animals and humans (both directions).

2. Why are carriers are dangerous reservoirs of infection? (mention 4 causes)

* They show no clinical symptoms, so they appear healthy.
* Their contacts are unaware of the risk, so precautions are not taken.
* They cannot always be managed effectively (e.g. a typhoid carrier preparing food.)
* They are more numerous than clinical cases and move freely in the community spreading infection.

3. Enumerate the three-mode biological mode of transmission by vectors and mention one example for each

* Prapagative type "only multiplication": plague bacilli in rat flea, yellow fever in female Aedes mosquito.
* Cyclapropagative type "multiplication and development": malaria in female anopheles mosquito.
* Cyclodevelopmental "only development": microfilariae ⟶ larval stage in female culex mosquito.

4. Mention Factors Affecting Outcome of Infection

* Pathogenicity and virulence of the micro-organism
* Antigenic power of the micro-organism
* Duration of infectious state (period of communicability)
* Ease of communicability of the disease

5. Mention 4 measures for Specific protection

* Immunization against vaccine preventable diseases
* Silver nitrate or pencillin eye drops against ophthalmia neonatorum
* Condom against AIDS
* Use of specific nutrients (Vitamin A against nutritional blindness)

6. What is The different aspects of rehabilitation are?

* Vocational rehabilitation (Restoration of earning capacity)
* Social rehabilitation (Restoration of relation in the society)
* psychological (Restoration of personal dignity and confidence)

7. What are Strict contraindications for vaccines?

* Very ill children who need hospital admission
* BCG is contraindication in AIDS and immune deficiency diseases
* pertussis component of DPT in :- * children with neurological disorders
* Severe reactions to previous DPT * Children over the age of five years

8. Mention for Causes of vaccination failure

* Heat : polio, measles, MMR, vaccines * Sunlight : BCG, DPT
* Chemicals :- Disinfecting and antiseptics, including spirits, are damaging to vaccine
* Time : All vaccines lose their potency after certain time
* Reconstitution : Reconstituted vaccines like BCG, measles, MMR are damaged after few hours from reconstitution

9. Predisposing factors of Health care-associated infections:

* Drug resistance in organisms
* overcrowding of the hospitals
* poor hospital environment
* Vulnerable groups : TB, diabetes, anemia, elderly, post-surgery, immunosuppressed

10. How can Droplet mode of transmission be prevented in Health care-associated spread?

* Avoid overcrowding
* Maintain 12-ft spacing between beds
* Ensure adequate lighting
* provide good ventilation, especially cross-ventilation

11. Enumerate Non-modifiable risk factors of non-communicable disease

* Age
* Sex
* Genetics factors
* Ethnicity

12. Enumerate Modifiable risk factors of non-communicable disease

* Tobacco
* Insufficient physical activity
* Harmful use of alcohol
* unhealthy diet :- * High salt * Saturated and trans fats
* obesity
* Raised blood pressure

13.Measures taken toward the contact of TB

* PPD testing → All household members and close contacts
* if PPD negative → repeat skin test in 2-3 month after exposure ends.
* if PPD positive → CXR mandatory → Continue preventive treatment for high risk groups (children ≤ 5 years old, HIV+, Close contacts) until repeated PPD is negative

14.Compare curative and preventive regimens of tuberculosis by listing the differences for each of the following features: Purpose, Target Group, Drugs Used, Duration

Regimens	Curative	Preventive
purpose	Cure the patient and stop transmission	prevent progression from latent infection to active TB
Target group	Confirmed TB patient (pulmonary or extra-pulmonary)	close contacts, children, HIV+, TST+ but no active disease
Drug used	Isoniazid, Rifampicin, pyrazinamide, Ethambutol	Isoniazid (INH)
Duration	usually 6 months (2 months intensive + 4 months Continuation)	usually 6-9 months

15.List the key physical environmental determinants of health."

* water and sanitation * Safe house
* clean air * Healthy workplace

16.Identify non-modifiable determinants of health."

* Age
* Sex
* Genetic factors
* Ethinicity

17.List the preventive measures that should be implemented for meningitis patient's close contacts & Outline the treatment plan for the patient.

* prevention :- . Adequate ventilation in crowded settings
. Health education and hygiene awareness
. Surveillance for early outbreak detection
* Treatment :- . First line :- pencillin
. close contact :- Chemoprophylaxis
. Carrier Eradication :- Rifampcin

18.List the age groups most susceptible to meningococcal meningitis.& Mention key environmental and social risk factors.

* most susceptible :- * All ages susceptible * Highest risk : infants 3-12 months
* Adolescent and young adults also vulnerable

* environment risk factors :- * Dry seasons * dust exposure * humidity fluctuations.
* Cold seasonal conditions

* Social risk factors :- * overcrowding * poor ventilation * limited medical access

19.In which situations is mass chemoprophylaxis recommended for meningococcal meningitis?

* It is only for closed, supervised communities ; e.g. military camps

20.Why is mass chemoprophylaxis not routinely recommended in meningococcal meningitis control?

* Rapid spread (because of high carrier rate) makes mass treatment impractical
* Risk of antibiotic resistance
* unnecessary exposure to drug side effects (it may expose many healthy people to antibiotics).
* Targeted prophylaxis is more effective

21.Answer the following questions:

a. Define leprosy elimination as a public health problem according to WHO.

* A prevelance of less than 1 case per 10,000 population

b.State the causative agent and reservoir of leprosy.

* Agent : Mycobacterium leprae
* Reservoir : Humans

c.Mention the main mode of transmission of leprosy.

* main mode of transmission: Respiratory droplets

d.What is the average incubation period of leprosy? What is meant by multidrug therapy (MDT) in leprosy control?

* Average incubation period → approximately 5 years
* (MDT): it's combination of drugs (Rifampicin, clofazimine, and Dapsone) used to prevent the development of drug resistance

e.State the recommended MDT regimen for paucibacillary leprosy

*for paucibacillary leprosy: Compination of:- Rifampicin and Dapsone for 6 months

f.State the recommended MDT regimen for multibacillary leprosy

*for multibacillary leprosy: Compination of:- Rifampicin, Dapsone and Clofazimine for 24 months

g.What vaccine provides partial protection against leprosy?

*BCG Vaccine

h.For how long should household contacts of a leprosy patient be examined?

*frequency:- Annually for at least 5 years (after the last exposure)
*purpose:- To ensure early detection of new cases among close contacts

i.Mention two important epidemiological features of leprosy transmission.

*Transmission occurs through prolonged close contact with an untreated case

*spread mainly by respiratory droplets from the nose and mouth

j.List three primary prevention measures for leprosy. What is the role of early case detection in leprosy control?

A. *Health education
*BCG Vaccine
*Early case detection, and prompt MDT

B. *Role of early detection:-
1-prevent disability 2-reduce transmission

k.Mention two public health problems associated with delayed diagnosis of leprosy.

* permanent physical disability
* Social Stigma

l.Why is monotherapy not recommended in the treatment of leprosy?

* unethical due to inevitable drug resistance

Which statement best reflects the dynamic nature of the health spectrum?

A. Health is a fixed state determined only by genetics.

B. Individuals move continuously between different levels of wellness and illness.

C. Health and disease are two separate unrelated entities.

D. The transition from health to illness is always abrupt.

Which individual demonstrates optimal social health?

A. A person with no disease but constant anxiety.
B. A person who maintains satisfying relationships and integrates well with society.
C. A person with no hospital admissions in the past year.
D. A person who meditates daily.

Which example best illustrates emotional health rather than mental health?

- A. Solving complex problems logically.
- B. Maintaining calm and self-control during stressful situations.
- C. Having high-level thinking and life philosophy.
- D. Absence of psychological disorders.

A community worker wants to improve population health by targeting the most powerful non-medical determinants. Which area should they prioritize?

- A. Hospital construction
- B. Income and social status
- C. Genetic screening
- D. Improving diagnostic technology

Which scenario best describes subjective well-being?

A. Blood pressure measurement
B. Clinical diagnosis of illness
C. A person reporting life satisfaction and happiness
D. A physician noting physical signs

A factory worker exposed to toxic chemicals develops illness. According to the epidemiological triad, which component is represented by the chemicals?

A. Host

B. Agent

C. Environment

D. Vector

Which period of disease is most dangerous for silent community transmission?

A. Convalescence
B. Illness period
C. Incubation period
D. D. Decline period

A patient states they feel ill although no pathology is detectable. This situation represents

A. Disease

B. Illness

C. Sickness

D. Recovery

Which determinant explains why well-educated individuals often have better health outcomes?

A. They have stronger genetics.

B. Education reduces stress and increases self-confidence.

C. Education eliminates environmental hazards.

D. Education directly increases immunity.

A patient recovering from infection starts feeling better but becomes vulnerable to secondary infections. Which disease stage is this?

A. Incubation period
B. Prodromal stage
C. Illness stage
D. Decline period

1. A surveillance team reports that carrier rates in a community increased from 8% to 32% over two months during the dry season. What is the most important next step?

- A. Initiate mass chemoprophylaxis for the entire region
- B. Intensify surveillance because this rise suggests early epidemic development
- C. Begin vaccination of only symptomatic individuals
- D. Close all schools and public transportation immediately

A meningitis outbreak is detected in a military training camp with rapidly rising carrier rates. Why is mass chemoprophylaxis indicated in this setting but NOT in general community outbreaks?

A. Military populations have weaker immunity
B. Vaccination fully replaces prophylaxis in general communities
C. Community populations do not respond to antibiotics
D. Military settings qualify as closed, supervised communities with high-density exposure, making mass prophylaxis effective and logistically feasible

In a suburban school, one case of meningococcal meningitis is confirmed. Carrier screening shows no evidence of widespread transmission. Which preventive action is most appropriate, and why?

A. Mass chemoprophylaxis because any school-based case signals community spread

B. Chemoprophylaxis for close classroom contacts only because mass prophylaxis is reserved for closed, supervised settings

C. Mass vaccination immediately because vaccines act faster than antibiotics

D. No prophylaxis required because children have strong natural immunity

meningitis post test

Points: 7/9

1. Name

[redacted]

✓ Correct 1/1 Points

2. **During an outbreak investigation, most cases occur in overcrowded dormitories with poor airflow. What does this pattern indicate epidemiologically?**

- (selected) A. Environmental factors are the primary drivers of transmission ✓
- B. Transmission is mostly via fomites on shared surfaces
- C. Low carrier rates are responsible for the outbreak
- D. Water contamination is the most likely source

✗ Incorrect 0/1 Points

3. **A confirmed meningococcal case receives appropriate antibiotics. After how long is the patient generally considered non-infectious, and why is this critical for control?**

- A. After 24 hours; reduces further droplet transmission ✓
- B. After 7 days; ensures total bacterial eradication
- (selected) C. After completion of full antibiotic course; prevents reinfection
- D. Only after repeat CSF cultures; confirms cure
- Option 2

✓ Correct 1/1 Points

4. **In a population with a steady carrier rate of 35%, what does this level most likely reflect in terms of epidemic potential?**

- A. The population has surpassed the epidemic threshold
- B. The level is at the upper end of inter epidemic baseline and requires monitoring ✓
- C. The rate suggests no risk of transmission
- D. The population is immunized sufficiently

✓ Correct 1/1 Points

5. **A school reports one confirmed case of meningococcal meningitis. Which control measure provides the highest impact in preventing further spread?**

- (selected) A. Immediate ventilation improvements and chemoprophylaxis for close contacts ✓
- B. Mass screening of all students with nasopharyngeal swabs
- C. Isolation of the case for 14 days
- D. Mandatory vaccination for all adults in the region

✓ Correct 1/1 Points

6. **In a military camp outbreak, mass chemoprophylaxis is recommended. Why is this approach appropriate in this particular setting?**

- (selected) A. Military camps qualify as closed, medically supervised populations ✓
- B. Soldiers have natural immunity due to prior exposures
- C. The disease cannot spread outside the camp
- D. Vaccination is ineffective in this age group

✓ Correct 1/1 Points

7. **A community health officer must choose between polysaccharide and conjugate vaccines for outbreak control. Which should be prioritized and why?**

- A. Conjugate vaccine; provides broader serogroup coverage and better herd immunity ✓
- B. Polysaccharide vaccine; gives lifelong immunity
- C. Polysaccharide vaccine; works best in children under age two
- D. Conjugate vaccine; requires multiple booster doses during epidemics
- Option 2

✓ Correct 1/1 Points

8. **Rising meningococcal carrier rates are detected, but no clinical cases have appeared yet. Which public health action is most justified?**

- (selected) A. Strengthen surveillance and environmental interventions ✓
- B. Begin mass vaccination immediately
- C. Initiate chemoprophylaxis for all community members
- D. Declare an epidemic and close public institutions

✗ Incorrect 0/1 Points

9. **A close-contact group is given rifampicin for chemoprophylaxis. What is the primary rationale behind using chemoprophylaxis instead of vaccination alone?**

- A. Vaccination does not work quickly enough to prevent immediate spread ✓
- B. Chemoprophylaxis permanently eliminates carrier states
- (selected) C. Chemoprophylaxis is more effective than vaccination long term
- D. Vaccination cannot be used during outbreaks
- Option 2

✓ Correct 1/1 Points

10. **A meningitis outbreak occurs in a refugee shelter. Which combination of actions will most effectively reduce transmission?**

- (selected) A. Improve ventilation + reduce crowding + prioritize chemoprophylaxis for close contacts ✓
- B. Isolate only symptomatic individuals and avoid environmental changes
- C. Perform mass CSF screening for all residents
- D. Provide polysaccharide vaccines to children only

"إذا غامرت في
شرف مروم فلا
تقنع بما دون النجوم"

A rural district reports a leprosy prevalence of 0.8 per 10,000 population but continues to detect high numbers of new multibacillary (MB) cases annually. Which interpretation best explains the situation?

A. The district has successfully eliminated leprosy as a public health problem and transmission has stopped.

B. Low prevalence means the district should stop active surveillance and focus only on treatment.

C. The low prevalence reflects effective MDT outcomes, but the high MB detection indicates ongoing transmission.

D. The high MB detection indicates that leprosy is becoming more contagious over time.

A family member has lived with an untreated MB leprosy patient for several years. Which factor most strongly determines whether the family member will develop clinical disease?

- A. Duration of exposure only
- B. The virulence of Mycobacterium leprae
- C. Level of the household contact's cell-mediated immunity
- D. Frequency of disinfection in the home

A health team must choose the most effective strategy to rapidly reduce community infectiousness in an endemic area. Which action will produce the fastest impact?

- A. Expanding BCG vaccination to all school-aged children
- B. Treating all confirmed MB cases with rifampicin-containing MDT immediately
- C. Annual screening of contacts for 5 years
- D. Hospitalizing all lepromatous patients for isolation

During a civil conflict, many patients default from MDT for months. What is the most likely epidemiological consequence?

- A. Sharp decrease in new cases due to reduced mobility
- B. Increase in drug resistance and rising community transmission
- C. Fewer MB cases because untreated disease becomes self-limiting
- D. Reduced nerve damage due to interruption of therapy

A newly diagnosed MB patient starts MDT today. The family asks when the patient stops being infectious. Based on the document, which response is most accurate?

- A. The patient remains infectious for at least 6 months after starting MDT.
- B. Infectiousness stops only after completing the full 24-month MB regimen.
- C. Infectiousness is markedly reduced within 3 days of receiving rifampicin.
- D. The patient is noninfectious immediately after the first dose of dapsone alone.

A 10-YEAR-OLD BOY IN A BOARDING SCHOOL DEVELOPS FEVER, HEADACHE, VOMITING AND NECK STIFFNESS. TWO MORE STUDENTS IN THE SAME DORMITORY DEVELOP SIMILAR SYMPTOMS WITHIN DAYS. LABORATORY TESTS CONFIRM NEISSERIA MENINGITIDIS.

- 1. Mode of transmission?
- 2. Why did the outbreak spread?
- 3. Public health control measures?

- Transmission:
- • Respiratory droplets

- Reasons for Spread:
- • Crowded dormitory
- • Close contact between students
- • Sharing utensils

- Control Measures:
- • Isolation and treatment of cases
- • Chemoprophylaxis for close contacts
- • Active surveillance
- • Health education
- • Vaccination in high-risk settings